

## Day 4 — Equinoxes — Viṣuva

**Module:** Movement of the Sun · **Band:** Senior · **Time:** 45 min

### Learning objective

By the end of this lesson, students will: identify spring and autumn equinoxes (day = night).

### Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

### Vocabulary introduced today

- *saṁkrānti* (IAST: saṁkrānti; lit. “Sun’s crossing from one Rāśi to the next”)

### Lesson flow

#### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

#### Core (20 min)

1. **Define equinox.** *Viṣuva* in Sanskrit. The two days each year (Mar 21 ± 1 and Sep 22 ± 1) when day length equals night length everywhere on Earth.
2. **Why?** At equinox, Earth’s axis is perpendicular to the Sun-Earth line. Sunlight falls equally on both hemispheres. The Sun rises due-east and sets due-west.
3. **Source.** Sūrya-siddhānta names the *viṣuva* points. Varāhamihira (*Brhat Samhitā*) discusses them in calendar context. The Vernal Equinox is the traditional anchor of the *sāyana* (tropical) zodiac.
4. **Cross-cultural.** The Roman / Gregorian calendar implicitly aligns to the equinoxes. Easter is computed by lunar-solar reckoning anchored to the Vernal Equinox. Many cultures celebrate the equinoxes — Holi (~Phālguna pūrṇimā, near Vernal Equinox), Mabon (NH autumn).
5. **Observation activity.** Each student notes the sunrise direction this week (or projects a stellarium app). Compare to expected (near-due-east if we’re near an equinox).

#### Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Equinoxes — Viṣuva.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

### Homework

(See `homework.md` for today’s task.)

### Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

### Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).